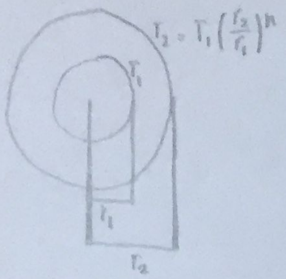
CHE 312 Quiz 4 Eric Huerta

$\frac{10}{10}$

1.




$$\left(\frac{T}{T_r}\right) = \left(\frac{r}{r_1}\right)^n$$

$$\left(\frac{D_{AB}}{D_{AB,1}}\right) = \left(\frac{T}{T_1}\right)^{3/2}$$

balance

$$\frac{d}{dr}(r^2 N_{Ar}) = 0$$

$$\frac{d}{dr}\left(r^2 \frac{cD_{AB}}{1-x_A}\frac{dx_A}{dr}\right) = 0 \rightarrow \text{must be integrated twice}$$

① $\int d\left(r^2 \frac{cD_{AB}}{1-x_A}\frac{dx_A}{dr}\right) = \int 0\,dr$

② $\frac{dx_A}{dr} = \frac{\int 0\,dr}{r^2}\cdot\frac{1-x_A}{cD_{AB}}$

$$\int dx_A = \int\left(\frac{\int 0\,dr}{r^2}\cdot\frac{1-x_A}{cD_{AB}}\right)dr$$

eqn: $\left(\frac{1-x_A}{1-x_{A1}}\right) = \left(\frac{1-x_{A2}}{1-x_{A1}}\right)^{\frac{((1/r_1)-(1/r))}{((1/r_1)-(1/r_2))}}$

$D_{AB}$ is a function of $r$; write it in terms of $r_1$

$$W_A = 4\pi r_1^2 N_{Ar}\big|_{r=r_1} = \frac{4\pi c D_{AB}}{(1/r_1)-(1/r_2)}\ln\left(\frac{1-x_{A2}}{1-x_{A1}}\right)$$

$$c = \frac{P}{RT}$$

$$\frac{d}{dr}\left(r^2 \frac{PD_{AB,1}/RT_1}{1-x_A}\left(\frac{r}{r_1}\right)^{n/2}\frac{dx_A}{dr}\right) = 0 \rightarrow \text{must be integrated twice}$$

① $\int d\left(r^2 \frac{PD_{AB,1}/RT_1}{1-x_A}\left(\frac{r}{r_1}\right)^{n/2}\frac{dx_A}{dr}\right) \int 0\,dr$

② $\frac{dx_A}{dr} = \int\frac{\int 0\,dr}{r^2}\cdot\frac{1-x_A}{PD_{AB,1}/RT\left(\frac{r}{r_1}\right)^{n/2}}$

$$W_A = 4\pi r_1^2 N_{Ar}\big|_{r=r_1} = \frac{4\pi\left(PD_{AB,1}/RT_1\right)[1+(n/2)]}{\left[(1/r_1)^{1+(n/2)} - (1/r_2)^{1+(n/2)}\right]r_1^{n/2}}\ln\left(\frac{1-x_{A2}}{1-x_{A1}}\right)$$